

ENERGY DATA HACKDAYS 2026

Energy Fingerprints

Make the invisible energy system visible.

From 15-minute smart-meter data to explainable household asset intelligence.

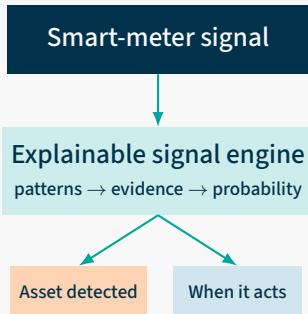
newspaper

AEW Energie challenge

The grid sees one number. We reveal the causes.

A household's net load hides the assets that shape flexibility, infrastructure demand, and solar potential.

- 4 years of anonymised, 15-minute readings
- ~90k households; ~1k labelled customers
- Privacy-preserving: no appliance sensors, no customer intervention



One interpretable workflow, five asset lenses



EV

Sharp switch-on steps in grid import.

Battery

Export displaced past solar noon.

Heat pump

Cold-weather and persistent heating signatures.

Other devices

Repeated distinctive load motifs.

Rooftop PV

Sunny-day midday import depression.

TBD: demo evidence view

EV: the model, and the rule it must beat

E22 LIGHTGBM • HELD-OUT TEST

45 features, threshold frozen before the test was opened. Test (67 properties, 15 EV-labelled): **ROC-AUC 0.67**. Detects EV *households*, not charging events.

RULE BASELINE

Flag EV if grid import ever steps up by ≥ 9 kW within one quarter-hour.

ROC-AUC 0.686 F1 0.50 n = 342

Threshold selected out-of-fold: 9.2 kW in $\sim 100/100$ grouped folds.

- The full plateau detector — band, duration, edge — scores 0.688: $\Delta\text{AUC} -0.002 [-0.033, +0.029]$ over one line of $\max(\Delta P)$.
- **$\sim 75\%$ of ≥ 9 kW ramp days fall in Nov–Mar for EV owners and non-owners.** Much of what every method here reads is winter heating, not charging.

LightGBM • held-out test

Actual cohort	Flagged	Not flagged
EV-labelled (GIGI)	10	5
Unlabelled properties	20	32

$$p \geq 0.392; n = 67.$$

Rule • 342 labelled samples

Actual cohort	Flagged	Not flagged
EV-labelled (GIGI)	49	23
Unlabelled properties	75	195

Precision 0.40 is a *floor*.

Heat-pump boiler: a physical detector, not a classifier

WPB DETECTOR • LABEL-FREE

The detector identifies heat-pump water-heater candidates from persistent plateaus in 15-minute grid-import data.

- p10 daily envelope: an event must recur on at least 90% of days, rejecting irregular loads such as part-time EV charging.
- Plateau band: 0.25–1.2 kW for 1.5–9 h; it must recur in both January and July.
- Population sweep: 838 of 75,094 households flagged (1.1%) as candidates.
- 4 labelled positives and 0 confirmed negatives: no confusion matrix or accuracy estimate.

Counterfactual recovery test

79.6%

Recovery at 0.3
kW injected

94.2%

Recovery at 0.5
kW injected

Synthetic events test sensitivity only;
they do not establish precision.

Rooftop PV: SolarPrint reads the solar imprint in demand

HISTGRADIENTBOOSTING · 5-FOLD OOF

23 import-shape features; feed-in is used offline for silver labels, never as a feature. **GIGI ROC-AUC 0.92.**

RULE BASELINE

Flag PV if the meter ever exported. You cannot inject what you do not generate.

F1 0.920 recall 0.922 n = 342

Zero tuned constants. A 0.2 kW deadband changed 0 of 342 calls, so it was removed.

- Import-only variant — midday draw ÷ night draw — scores **0.800** against a gradient-boosted model's 0.806 on the same samples.
- **The rule's 23 false positives mostly have PV:** median export 17.0 kWh/day vs 19.6 for labelled PV homes; true negatives export exactly zero.

SolarPrint · population

Actual cohort	Flagged	Not flagged
---------------	---------	-------------

GIGI PV		
(confirmed)	260	24
Strictly unlabelled	4.6k	65.6k

$p \geq 0.5$; silver excluded.

Rule · 342 labelled samples

Actual cohort	Flagged	Not flagged
---------------	---------	-------------

GIGI PV		
(confirmed)	259	22
Unlabelled properties	23	38

Different denominators — not comparable cell for cell.

Battery: a physical rule already reads storage

RULE BASELINE

Of the solar exported in the shoulder months, what share leaves the house *after* solar noon?

ROC-AUC 0.855 F1 0.892 n = 242

Flag a battery above 0.50. Nothing here is fitted to the labels.

- **Why 0.50.** The sun is symmetric about noon: a home *without* storage exports half before, half after. A battery eats the morning surplus and spills only once full. Non-battery homes measure 0.466, battery 0.561.
- **The trap we avoid.** 96% of batteries sit in a PV home, so `has_PV` alone scores 0.700 against battery. Every number here is measured **within PV homes**, where `has_PV` is worth 0.500.

Rule • 242 PV households

Actual cohort	Flagged	Not flagged
GIGI battery (confirmed)	170	20
Unlabelled PV homes	21	31

Share after noon ≥ 0.50 . This is the rule baseline — the trained PV-conditioned model is still to come.

The seasonal test

	No batt.	Battery
Jun–Aug	0.459	0.529
Mar/Apr/Sep/Oct	0.483	0.632

Not west-facing roofs. A battery saturates in high summer so the skew shrinks — and it does, by 0.103. Non-battery homes move 0.024. A fixed roof cannot change azimuth.

Other-device classifier: turn unknown load shapes into leads

OTHER DISTINCTIVE DEVICE

The model flags repeated, unusual load motifs for review, then groups them into candidate device classes as labels mature.

- Input: recurring event shape, power, duration, timing, and seasonality
- Output: device-class probability or “novel pattern” candidate
- Evidence: a small set of representative events a human can inspect

TBD: cluster of recurring appliance signatures

Evaluation placeholder

Actual cohort	Flagged	Not flagged
Asset present (TBD)	TBD	TBD
Asset absent (TBD)	TBD	TBD

Define the class and held-out split before reporting results.

Trust is a feature, not a footnote

Explainable

Every score
links to its
evidence.

“Why was this home
flagged?” has a visible
answer.

Honest

Validation is
held out and
limitations stay
visible.

We do not invent accuracy
where labels are missing.

Scalable

One workflow
can score the
full population.

Designed for multi-year,
15-minute data.

TBD: insert one anonymised household evidence card here

From meter data to better grid decisions.

Asset intelligence, activity windows, and evidence—at population scale.

Forecasting

Better demand expectations

Planning

Target infrastructure investment

Flexibility

Find the right homes for the right programme

TBD: closing ask / next experiment

Thank you